FLEET TEAM DIGEST

737MAX-FTD-34-23007

Issue Title : GPS RFI Signal Loss Event Data

Airplane Model	ATA	Minor Model(s)
737MAX	3458-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
767, 737NG, 747, 787, 777, 757, 737	9E991

Originated Date	Last Revised Date	Created On
10/17/2023	09/18/2026	08/17/2026

Estimated Completion Date	Next Update Date
12/30/2027	12/18/2026

Revision Description

Added information about the GLU-920/GLU-925 Date non-recovery into the Status Section.

Added GEM Service Letter Info to Interim Action Section.

Updated Milestones.

Description

Boeing is aware of the increasing number of dual GPS signal loss events occurring in the various fleets due to Radio Frequency Interference (RFI). This FTD provides information regarding the data that is being requested to perform an investigation, for cases where dual GPS signal loss events due to RFI are reported to Boeing.

Background

GPS satellite signal which provides the information needed for a GPS receiver to determine a Position/Velocity/Time (PVT) solution originates from space, and therefore GPS receivers may lose signal occasionally due to various unpredictable factors such as atmospheric disturbances, weather, solar activity, and intentional or unintentional signal interference. This results in a temporary inability of the receiver to calculate a valid PVT solution however, in most cases normal system operation will resume once proper GPS satellite signal is reacquired by the GPS receiver.

For cases where the GPS functionally is lost due to a hard fault within the hardware, the airplane has been designed to alert the crew of the failed hardware, so that the ground maintenance crew can be informed and the GPS related hardware can be inspected and replaced as necessary.

A number of GPS signal interference events have been reported in certain areas around Europe, the Mediterranean, and the Middle East which has prompted the European Union Aviation Safety Agency (EASA) to release Safety Information Bulletin (SIB) 2022-02R3, dated 05-JUL-2024, which warns operators of the specific areas where GPS signal interference may occur.

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Similarly the Federal Aviation Administration (FAA) has released SAFO 24002, dated 01-25-2024, to provide operators with information related to GPS disruptions.

Status

A number of airplane systems are reliant on valid GPS data, and a loss of GPS data can affect multiple downstream systems that utilize this data. Boeing has released model based Flight Operations Technical Bulletins (FOTB) and Operations Manual Bulletins (OMB), noted in the References Section, that provide information which may prove useful in airlines operations or airline training. As new symptoms are discovered, the FOTBs will be updated to capture any new effects.

Industry is aware of this evolving concern and is currently working to develop updated minimum operating specifications for the GPS receivers in order to improve the resiliency of the units, which is expected to be a solution for the distant future.

All GPS receivers currently deployed on Boeing airplanes can be affected by GPS RFI regardless of the supplier of the unit.

Airplane model specific symptoms and details for currently known GPS RFI effects, can be found in the FOTB for that specific model.

The most frequently reported symptoms that are commonly reported across the majority of the fleets are as follows:

- Non-recovery of Collins GLU-920/GLU-925/GLU-2100 MMR units - After experiencing prolonged GPS RFI automatic recovery of these units may not occur until a power cycle is performed and a Non-Volatile Memory (NVM) reset may be required. To resolve this an updated OPS software (phase 2.6) for the GLU-2100 MMR has been developed to improve non-recovery. Boeing is working to release a SB to allow installation of this software. Additionally, Collins has informed Boeing that the GLU-920/925 MMR models use older hardware components which are unable to handle an updated software that addresses GPS RFI, therefore no updates will be available for these model receivers.
  - Recently Boeing has received reports of cases where a power cycle does not restore GPS Date functionality on GLU-920/GLU-925 MMRs. Boeing and Collins are working to recreate epoch-shift condition in lab environment to determine if there is an alternate method to reset back to accurate date parameter without an NVM reset. Currently the unit will need to be sent to Collins for recovery, when this occurs.
- False EGPWS Pullup Alerts – GPS RFI may cause the EGPWS to trigger false alerts during or after leaving the RFI area. This phenomenon is known to occur for all GPS manufacturers, including Honeywell & older standalone units. The EGPWS supplier, Honeywell, has indicated that the unit will correctly re-calibrate upon landing as long as the horizontal GPS position has recovered. An updated software is being developed for the MKV-A units and airplanes with a MKV unit will need to upgrade to a MKV-A (ref Honeywell SIL).
- Momentary Lateral Deviation & ANP Growth - During a GPS RFI spoofing event, the FMS may intermittently switch Navigation Sources, causing a momentary lateral movement. The FMS selects the most accurate updating source that is valid and reasonable, and when GPS becomes unavailable or inaccurate this can cause the ANP to grow. The ANP will change based on the currently available sensors and should be considered valid.
- Emergency Locator Transmitter - Distress Tracker (ELT-DT) - Extended/repeated EGPWS Pull Up alerts may cause the ELT-DT to activate and ‘ELT ON’ to illuminate. This condition will resolve automatically once the alert has been canceled for a sufficient amount of time. Additionally, loss of GPS signal may trigger the ‘ELT SYS’ status message along with the associated maintenance messages: ‘ELT-DT’ Antenna Failed' and ‘ELT-DT has an Internal Fault’. The ELT-DT will continue operating normally and the last known valid location will be used if the unit is activated.

Additional information related to frequently asked questions are as follows:

- The Inertial Reference System (IRS) used on all Boeing airplanes outputs pure inertial data that is independently derived based on internal sensors. Pure inertial outputs are independent from and not reliant on GPS input. Airplane models that utilize additional hybrid GPS-IRS functionality will still have the pure inertial data available for FMS use. GPS RFI cannot cause IRS failures.
- There is currently no known mechanism for FMS failure due to a GPS RFI event.

As further reports of any new symptoms are discovered, Boeing will continue to catalog and analyze airplane level effects in support of developing future GPS RFI resilient systems.

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Interim Action

- Boeing has released updated model based FOTBs to provide the latest known information related to GPS signal interference and is currently collecting additional information for any new symptoms not already detailed in the FOTBs, in order to improve overall airplane system resilience to the loss of GPS data.

- Boeing has released the Operations Manual Bulletin (OMB) noted in the references section to provide guidance to flight crews when GPS RFI is experienced.

- Airplane Maintenance Manual (AMM) tasks related to the GPS have been reviewed and updated as needed to accomplish more through testing. Additionally, the Fault Isolation Manual (FIM) has been reviewed and updated as needed to add troubleshooting of all currently known scenarios related to the loss of GPS.

- Boeing has released the Maintenance Tip noted in the references section to inform maintenance crews of the currently known symptoms of GPS RFI and if it has been determined an airplane has flown through GPS RFI, the GPS System Test AMM Task should be accomplished to support recovery from possible residual effects and to verify proper GPS system operation.

- Updated GLU-2100 phase 2.6 operational software, which will mitigate non-recovery for known GPS RFI scenarios and introduce a simpler NVM reset procedure has received TSO approval and Boeing is currently working to certify and release a Boeing SB to allow installation of this software.

- Boeing is working with Honeywell to develop & deploy an updated MKV-A EGPWS software fix with improved logic to mitigate the false EGPWS pullup alerts for currently known GPS RFI triggers.

- Boeing developed a GPS Event Monitoring (GEM) Airline Modifiable Information (AMI) that includes functionality to detect when an airplane enters and exits a GPS RFI condition. Boeing released 737-SL-31-088 to provide further details regarding the installation and use of this software.

- Boeing is working with Collins on evaluating installation of a Collins GLU-2100 MMR Phase 3 software which will include spoofing detection capability.

- Although Boeing has released FOTBs that provide information and guidance related to currently known GPS RFI symptoms, due to regulatory rules it is unable to release a flight crew procedure to perform a Circuit Breaker (CB) reset during flight. However, Boeing is aware that some operators may be interested in developing their own procedure and working with their local regulatory agency to apply for limited maintenance authority, which would allow an in-flight CB cycle. Boeing is offering to perform a cursory review of any related operator procedure to accomplish this task and offers the information in attachment “737 In-Flight CB Cycling Considerations” to support this procedure development.

Final Action

TBD

Milestones

Note: As this is an industry wide concern involving various suppliers and working groups, milestones will be updated as necessary to support the various efforts needed to improve the impacted systems. Target dates are based upon currently available status and have risk of future revision.

Continue Working with Industry Standard Development and Industry Panels to develop GPS RFI Improvement Specifications:

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Ongoing

Catalog RFI Threats by Model, Phase of Flight, and Rate of Events: Ongoing

Release FOTB 737-23-01 (non-hybrid) - Latest Revision Dated: 25-FEB-2025 (Complete)

Release FOTB 737-24-01 (hybrid) - Latest Revision Dated: 25-FEB-2025 (Complete)

Release 737 MT 34-066 R3 - Latest Revision Dated: 30-JAN-2026 (Complete)

Release 737 OMB TBC-39 - Latest Revision Dated: 16-DEC-2024 (Complete)

Release 737-SL-31-088 - Latest Revision Dated: 11-MAY-2026 (Complete)

Release Boeing SB for installation of Collins GLU-2100 MMR Phase 2.6 Software: Target Q4 2026

Release Honeywell STC for MKV-A Software Update: Target Q1 2027

Obtain Collins GLU-2100 MMR Phase 3 Software TSO Approval - Target Q2 2027

Operator Action

Normal maintenance practices should be followed before submitting a report, in order to determine if airplane hardware, such as faulty antennas/cabling or loose/dirty connectors, are the cause for repeated dual GPS outages, or single side GPS outages, reoccurring on a specific tail. After receiving a report of an airplane flying through GPS RFI a check of the GPS functionality can be accomplished, in the absence of GPS RFI, by performing the GPS System Test, AMM task noted in the References Section of this article.

If GPS RFI is suspected for any flight along with any new symptoms which are not already called out in the FOTB, Boeing kindly requests a service request be sent in the Boeing Communication System (BCS) to inform us of the event. For any service request submitted related to a GPS RFI event, please complete the attached “GPS RFI Information Submission Form” questionnaire and provide with the service request submission.

GPS RFI events that are experienced along with already known symptoms called out in the FOTB do not need to be reported in an individual service request, however for tracking purposes, Boeing requests operators to submit the number of events reported over specific FIRs for each known symptom (as shown in the attached template – GPS RFI Event Tracking Sheet), on a monthly basis if possible. This data will support tracking of any pattern changes or evolution in the GPS RFI environment.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Other	FAA SAFO 24002		01/25/2024
Other	EASA Safety Information Bulletin (SIB) 2022-02	R3	07/05/2024
Flight Operations Technical Bulletin (FOTB)	737 23-01	R2	02/25/2025
Operations Manual Bulletin	TBC-39	New	12/16/2024

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(OMB)			
Fault Isolation Manual (FIM)	FIM Task 34-00-00-810-855		
Aircraft Maintenance Manual (AMM)	Task 34-58-00-730-801: GPS - System Test		
Maintenance Tip (MT)	737 MT 34-066	R3	01/30/2026
Service Information Letter (SIL)	Honeywell SIL D201312000067	1	08/31/2020
Flight Operations Technical Bulletin (FOTB)	737 24-01	R1	02/25/2025
Service Letter (SL)	737-SL-31-088		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message
- Flight Operations : Flight Operations

Attachments:

a) GPS_RFI_Event_Tracking_R0

b) 737 In-Flight CB Cycling Consideration

c) GPS RFI Information Submission Form

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